Which buildings waste energy?

Predicting building energy use with MATLAB, and turning predictions into a retrofit shortlist

NTU Learn to Hack: MATLAB Edition 2026 · Predicting Building Energy Consumption

github.com/JoshiiWahWah/ntu-matlab-hackathon-2026

THE PROBLEM

Not which buildings use the most, but which use more than they should

Buildings are major contributors to global energy use and carbon emissions.

Retrofits pay off most where a building uses far more than its type, size, age and climate explain.

We predict annual site Energy Use Intensity (kBtu/ft²/yr) for 75,757 building-years (WiDS 2022), entirely in MATLAB.

PHYSICS FRAMING

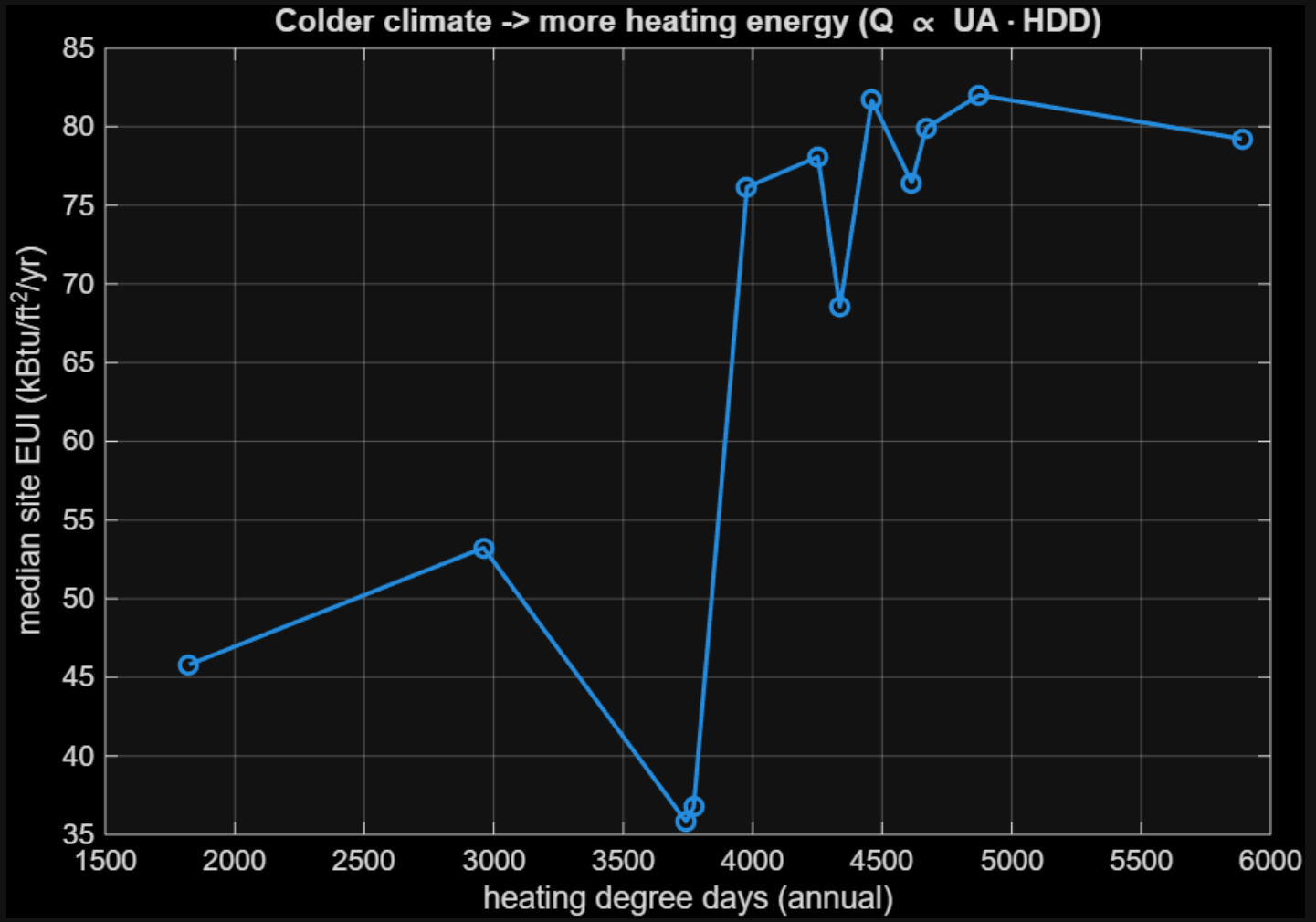
Energy balance: Q ≈ UA · ΔT summed over the year

Heating demand scales with heating degree days, cooling with cooling degree days.

Features we engineered:

winter & summer mean temperature · annual temperature range · total degree days · cooling share · log floor area · building age

Raw data: median EUI ≈ 36–53 in milder climates vs ≈ 68–82 in colder ones.



DATA MASTERY

Three findings that shaped the whole design

1 Distribution shift

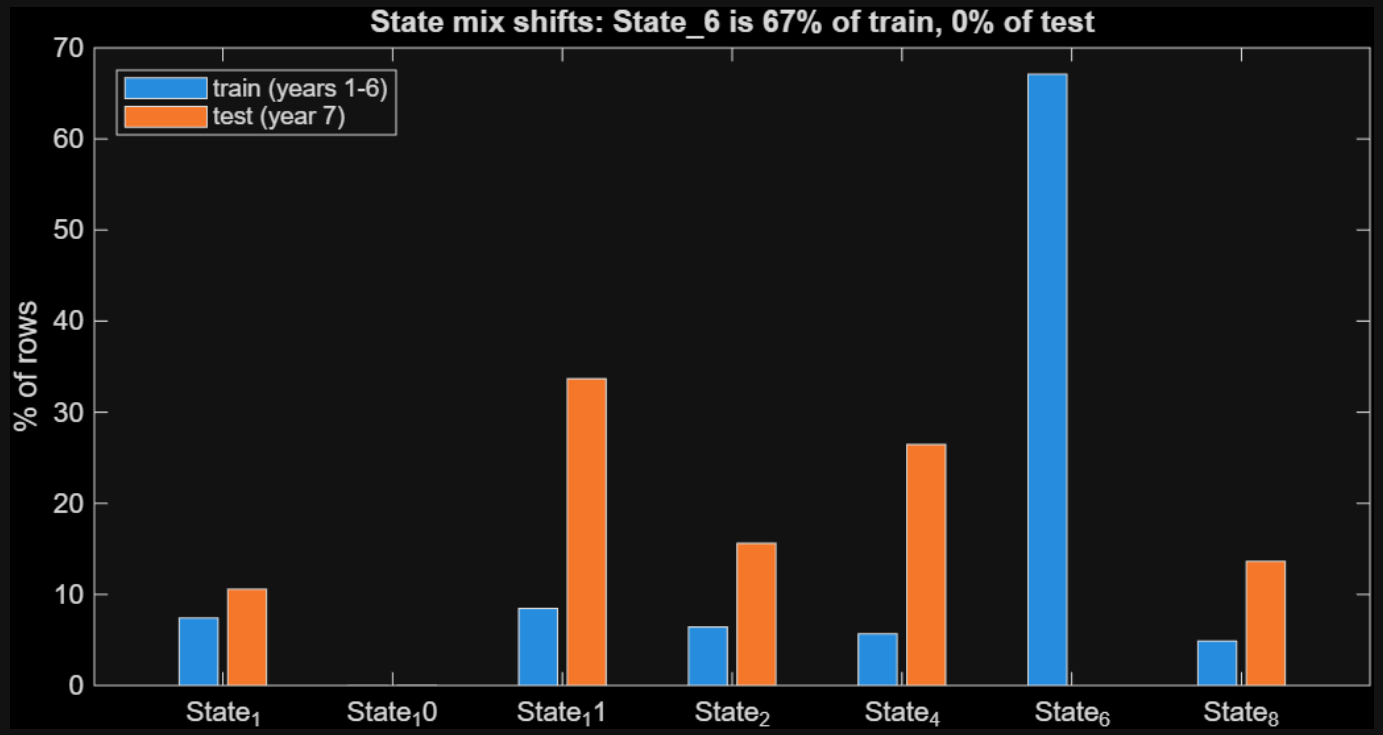
State_6 is 67.1% of train but 0% of test → we also report a test-like score.

2 Missingness shift

Wind/fog 56.0% missing in train vs 89.8% in test → dropped.

3 Hidden building identity

Reconstructed 63,907 buildings → past energy use as a feature, from earlier years only (unit-tested).



HONEST VALIDATION

Validate the way the task works: train on the past, predict an unseen year

Train: years 1–5

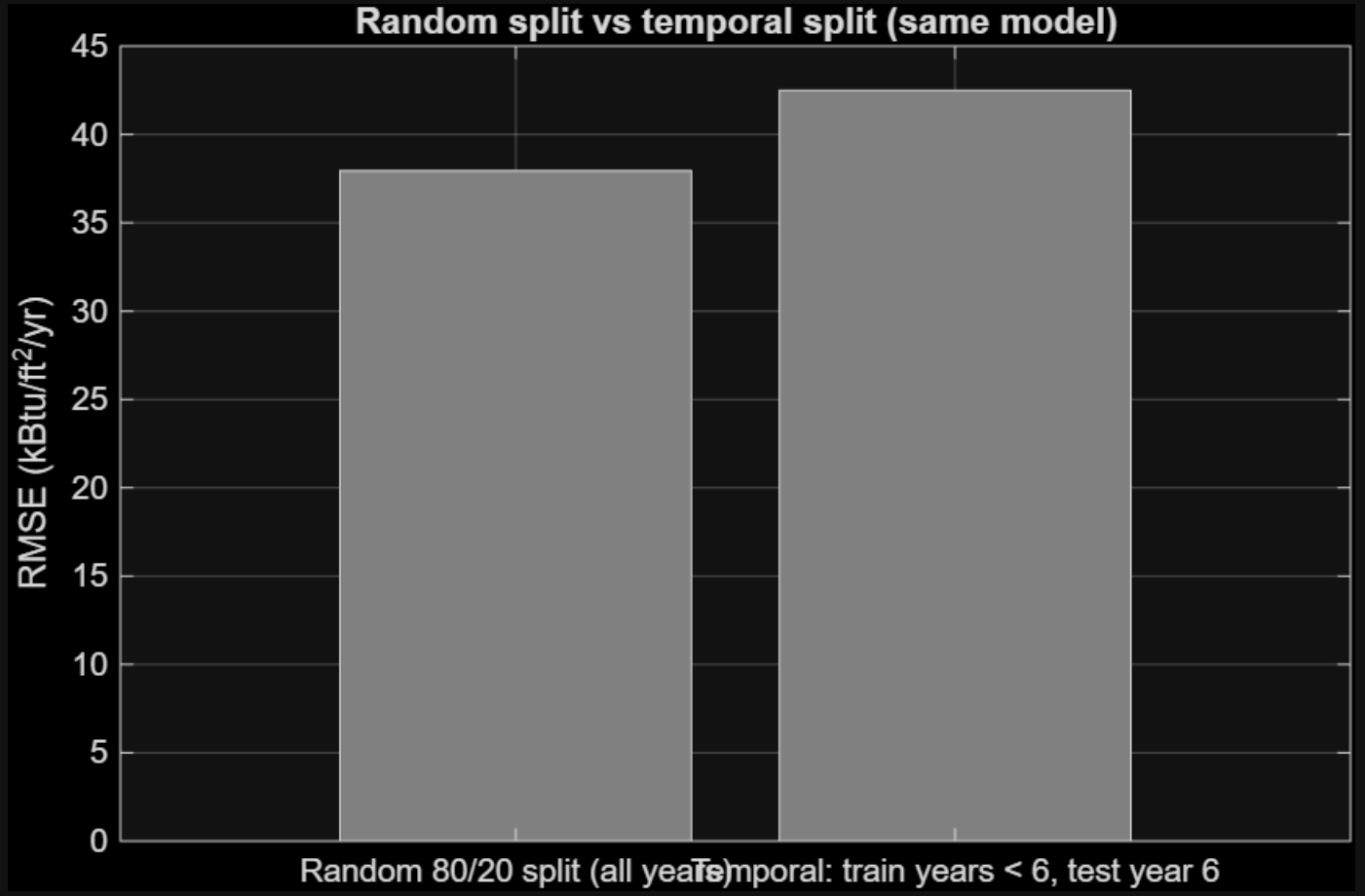
Validate: year 6 (test is year 7)

Same model, same features:

Random split 37.9

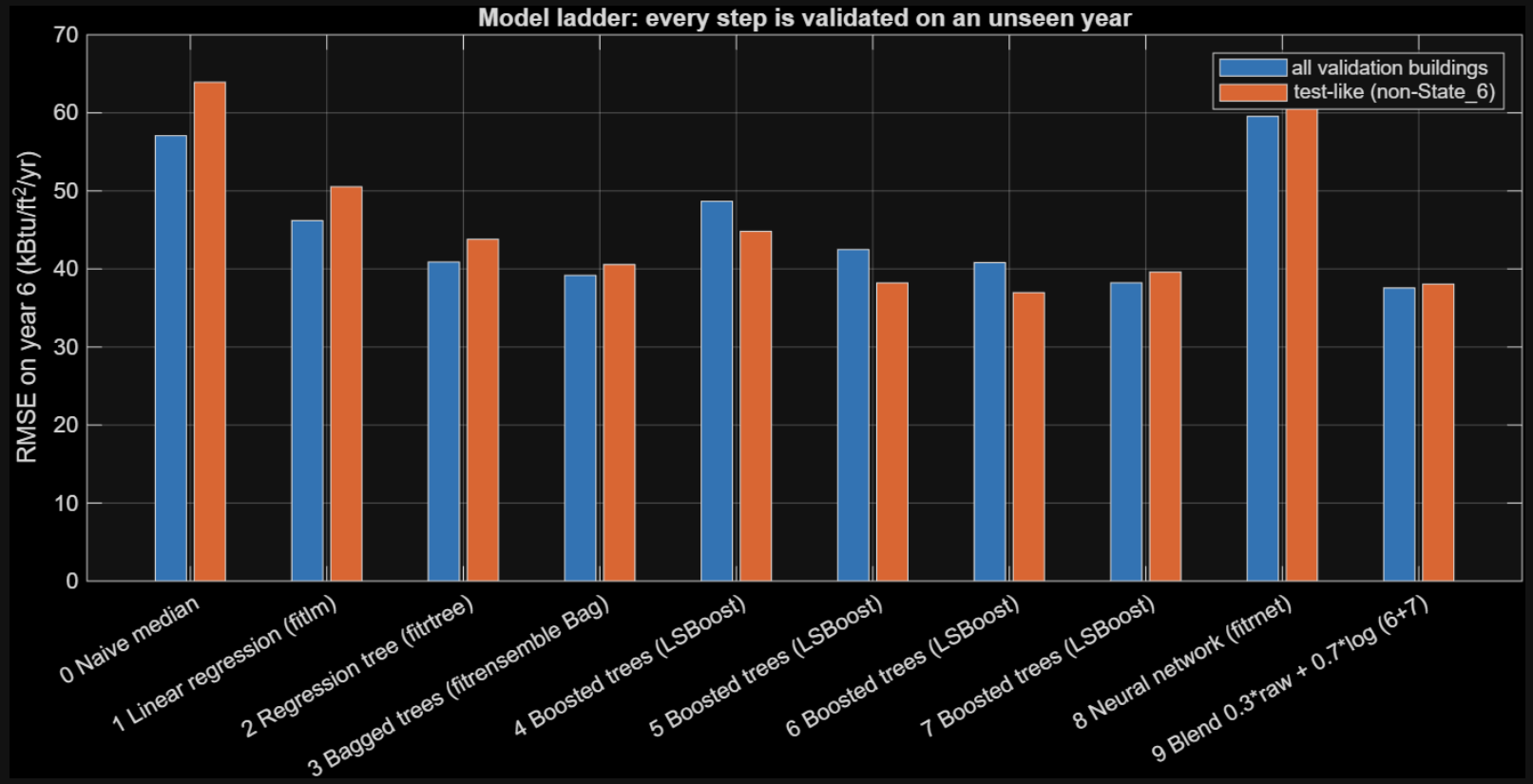
Temporal split 42.5

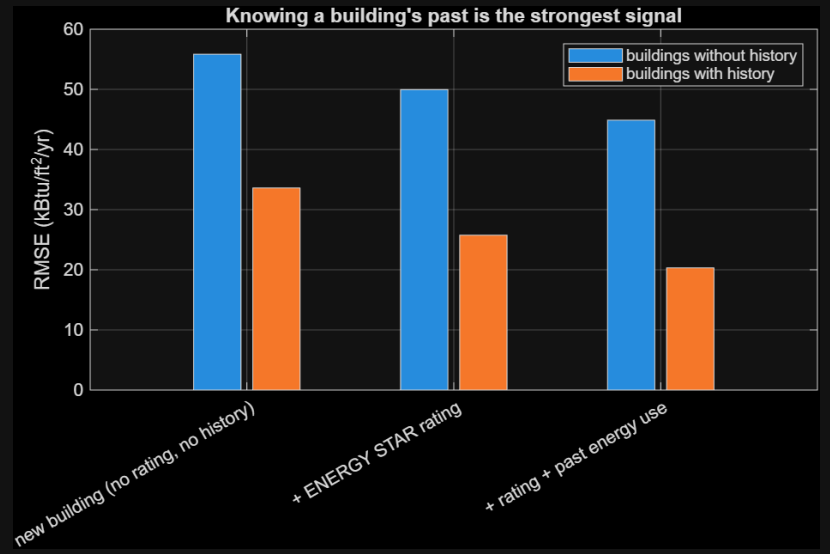
A random split flatters by ~12%: other years of the same building leak into training.



USE OF MACHINE LEARNING

10-model ladder: each rung answers one question





Linear 46.2 → tree 40.9: interactions matter. + ENERGY STAR rating, + building history, raw/log blend.

Best: blend of raw + log boosted trees

RMSE 37.6

−34% vs naive (57.1) · R^2 0.57

Buildings with history: RMSE 20.3

WHAT WE LEARNED THE HARD WAY

How MATLAB trees treat missing values

Without surrogate splits, a row missing the split variable stops at that node.

ENERGY STAR rating (35% missing) and history (~60% missing) were our strongest features, so those buildings were barely learned.

Fix: encode missing as an out-of-range sentinel (src/treeInputs.m) and keep explicit missing flags.

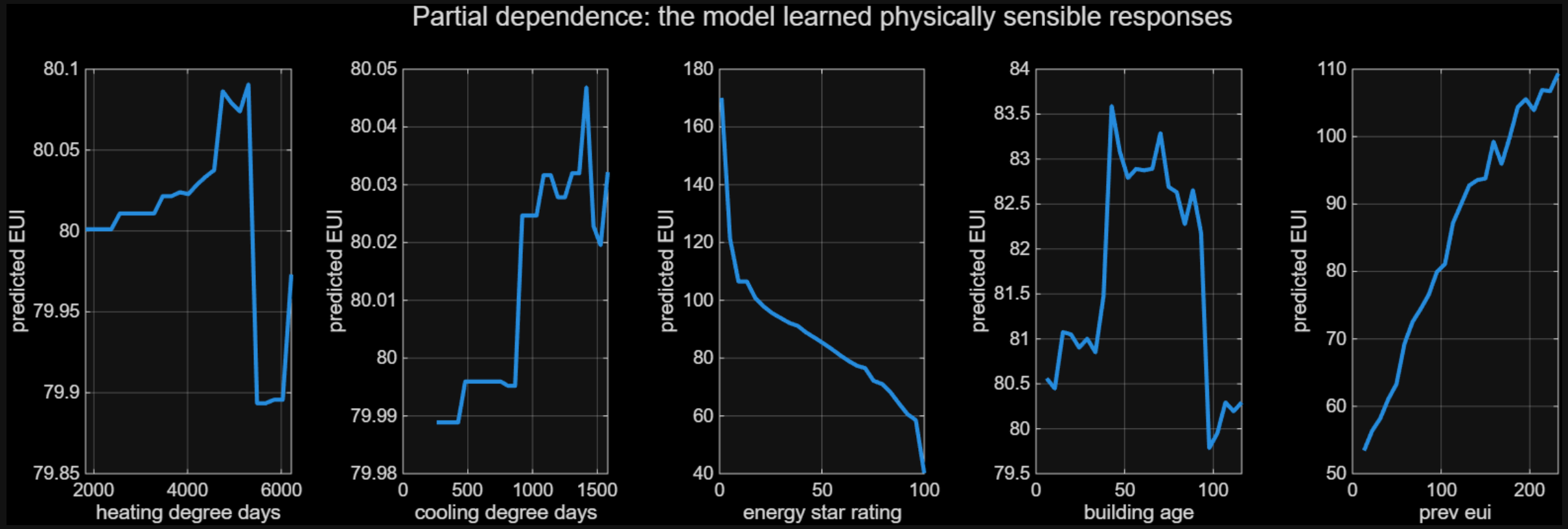
blend RMSE

46.8 → 37.6

first full run → after the fix

INTERPRETABILITY

What drives energy use



Higher ENERGY STAR score → lower EUI; past use predicts future use. Top drivers: facility type, ENERGY STAR rating, previous-year EUI, floor area.

Honest caveat: weather is recorded per state and year, so once the state is known the degree-day features add little.

IMPACT

From prediction to action: the retrofit shortlist

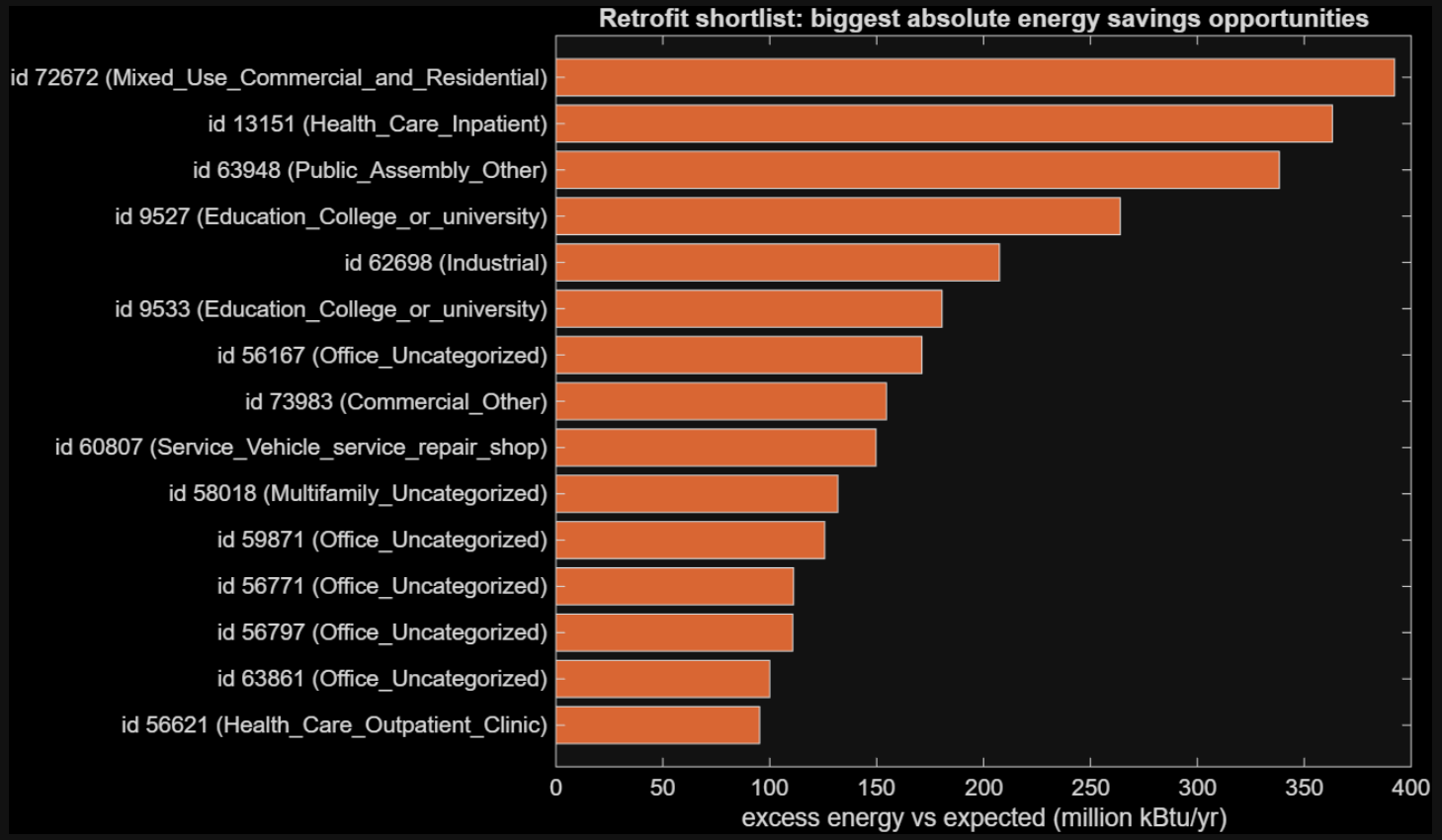
A cold-start model predicts what a typical building of the same type, size, age and climate should use.

Candidates: top 5% excess within each facility type, ranked by excess EUI × floor area.

1,123 buildings

1.55×10^{10} kBtu/yr above expected

5.3% of all energy used, concentrated in ~5% of buildings



ENGINEERING QUALITY

One command, reproducible, tested

>> main

Raw CSV → cleaning → features → 10-model ladder → validated submission + 12 figures

Modular functions in src/ with argument validation

matlab.unittest tests, including proof that history features are causal

Every number in the README is printed by MATLAB

Next steps

Monthly meter data and occupancy

Optimise which retrofits to fund under a budget

Per-building uncertainty

Thank you

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